

The entropy of a discrete random variable X with distribution p is defined as

$$\mathbb{H}(X) = -\mathbb{E}_{X \sim p} [\log p(X)] = -\sum_x p(x) \log p(x)$$

also denoted as $H(p)$. When the random variable X is continuous, the entropy is known as differential entropy. It gives a lower bound on the number of bits (if logarithm base 2 is used) needed to encode the outcomes of X using an optimal code.

0.0.1 See also

[Kullback-Leibler Divergence.tex]

[Cross Entropy]

0.0.2 References